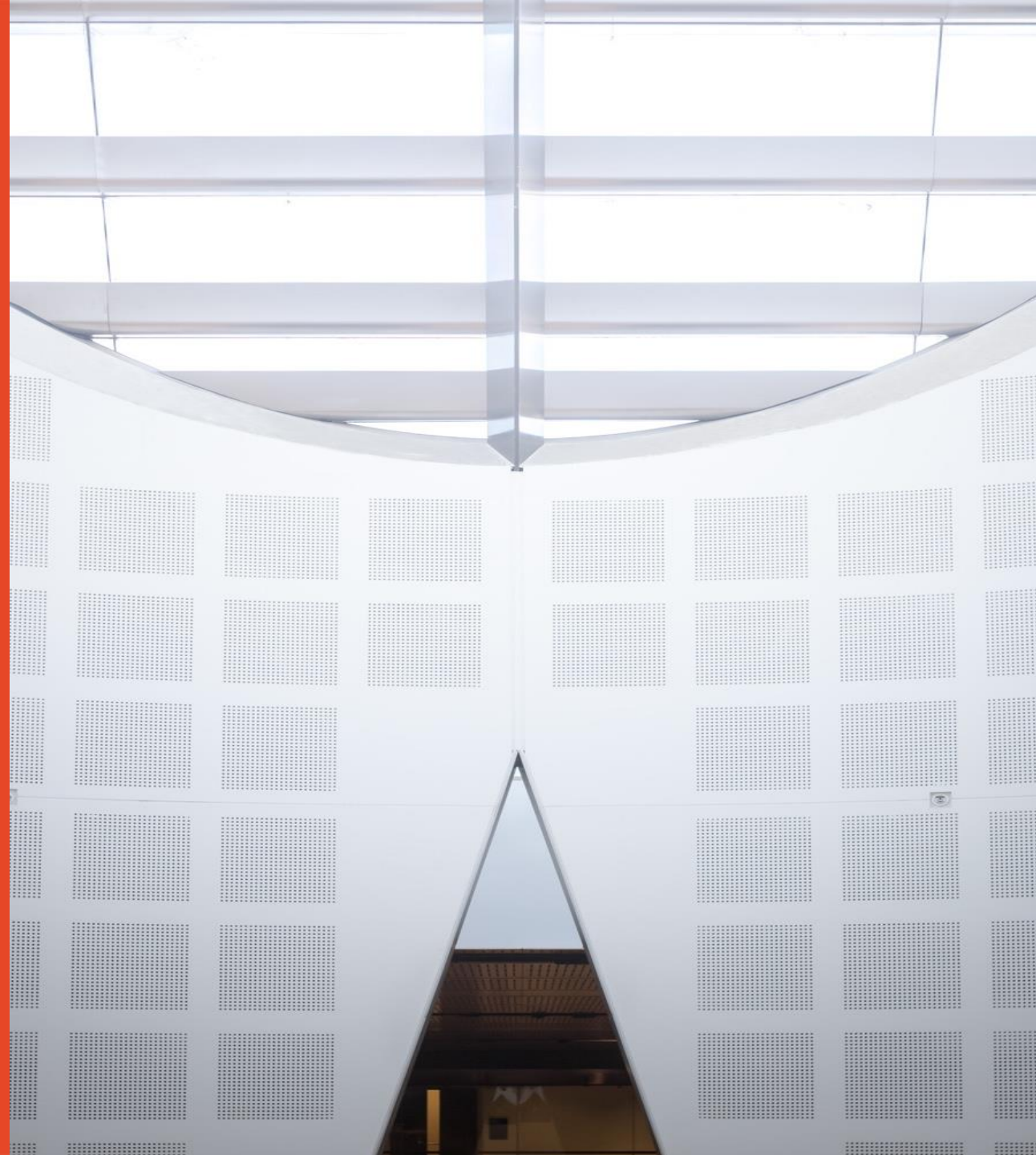


Solid State and Devices

Dr Marco Fronzi
School of Physics



THE UNIVERSITY OF
SYDNEY



- Application of the Schrödinger Equation to the Hydrogen Atom
- Solution of the Schrödinger Equation for Hydrogen
- Quantum Numbers
- Periodic Table

The atom of modern physics can be symbolized only through a partial differential equation in an abstract space of many dimensions. All its qualities are inferential; no material properties can be directly attributed to it. An understanding of the atomic world in that primary sensuous fashion...is impossible.

- Werner Heisenberg

Application of the Schrödinger Equation to the Hydrogen Atom

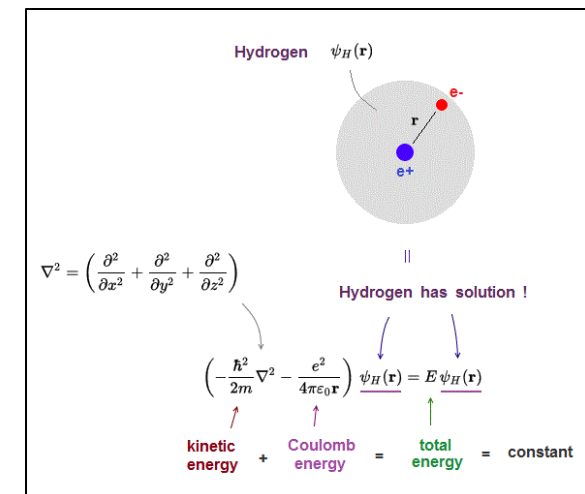
- The approximation of the potential energy of the electron-proton system is electrostatic:

- Rew
$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x) \psi(x) = E \psi(x),$$

- write the three-dimensional time-independent Schrödinger Equation.

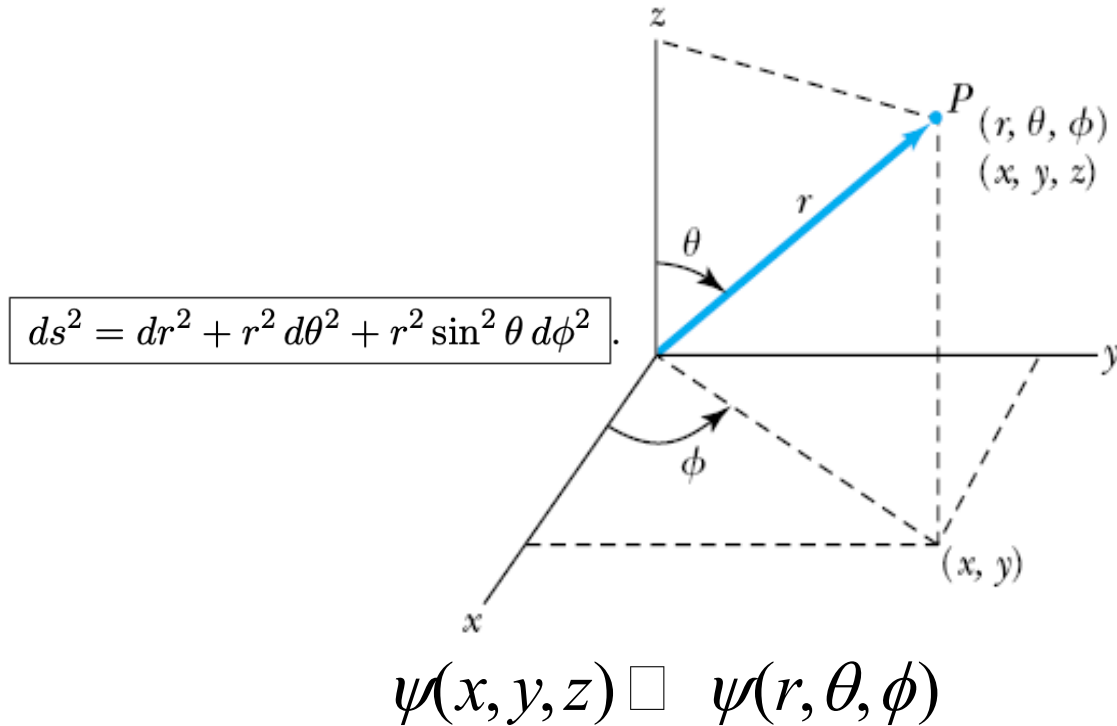
$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

- For Hydrogen-like atoms (He^+ or Li^{++})
- Replace e^2 with Ze^2 (Z is the atomic number).
- Use appropriate reduced mass μ .



Application of the Schrödinger Equation

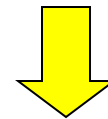
- The potential (central force) $V(r)$ depends on the distance r between the proton and electron.



$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \\ r &= \sqrt{x^2 + y^2 + z^2} \\ \theta &= \cos^{-1} \frac{z}{r} \text{ (Polar angle)} \\ \phi &= \tan^{-1} \frac{y}{x} \text{ (Azimuthal angle)} \end{aligned}$$

Transform to spherical polar coordinates because of the radial symmetry.

Insert the Coulomb potential into the transformed Schrödinger equation.



$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} + \frac{2\mu}{\hbar} (E - V(r)) \psi = 0$$

Solution of the Schrödinger Equation for Hydrogen

$$\psi(\mathbf{r}) = R(r) Y(\theta, \phi).$$

$$\psi(r, \theta, \phi) = R(r) f(\theta) g(\phi)$$

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}.$$

$$-\frac{\hbar^2}{2\mu} \left[\frac{1}{r^2 Y} \frac{\partial}{\partial r} \left(r^2 \frac{dR}{dr} \right) + \frac{1}{r^2 R} \left\{ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right\} \right] - \frac{Ze^2}{4\pi\epsilon_0 r} = E.$$

$$-\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2\mu r^2}{\hbar^2} \left(-\frac{Ze^2}{4\pi\epsilon_0 r} - E \right) = \frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right].$$

Solution of the Schrödinger Equation for Hydrogen

$$-\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2\mu r^2}{\hbar^2} \left(-\frac{Ze^2}{4\pi\epsilon_0 r} - E \right) = \lambda,$$

$$\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] = \lambda.$$

$$Y(\theta, \phi) = \Theta(\theta)\Phi(\phi).$$

Solution of the Schrödinger Equation for Hydrogen

$$\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] = \lambda.$$

$$Y(\theta, \phi) = \Theta(\theta)\Phi(\phi).$$

$$\frac{\sin \theta}{\Theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = \lambda \sin^2 \theta.$$

The first term depends only on θ , the second only on ϕ , hence each must equal a constant. Let this constant be $-m$

$$\frac{d^2 \Phi}{d\phi^2} + m^2 \Phi = 0.$$

$$\Phi_m(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}, \quad m \in \mathbb{Z}.$$

Solution of the Schrödinger Equation for Hydrogen

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) - \frac{m^2}{\sin^2 \theta} \Theta + \lambda \Theta = 0.$$

- Left and right side independent variables

$$(1 - x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[\lambda - \frac{m^2}{1 - x^2} \right] \Theta = 0.$$

$$Y_{\ell}^m(\theta, \phi) = N_{\ell m} P_{\ell}^m(\cos \theta) e^{im\phi},$$

$$N_{\ell m} = \sqrt{\frac{2\ell + 1}{4\pi} \frac{(\ell - m)!}{(\ell + m)!}}.$$

Solution of the Schrödinger Equation for Hydrogen

$$-\frac{1}{R(r)} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2\mu r^2}{\hbar^2} \left(-\frac{Ze^2}{4\pi\epsilon_0 r} - E \right) = \ell(\ell + 1).$$

- Left and right side independent variables

$$-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \left[\frac{\hbar^2 \ell(\ell + 1)}{2\mu r^2} - \frac{Ze^2}{4\pi\epsilon_0 r} \right] u(r) = E u(r).$$

$$R_{n\ell}(r) = \frac{2}{n^2} \left(\frac{Z}{a_0} \right)^{3/2} \sqrt{\frac{(n - \ell - 1)!}{(n + \ell)!}} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho).$$

$$E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2 \hbar^2} \frac{1}{n^2}, \quad n = 1, 2, 3, \dots$$

- Angular part $\Rightarrow$ spherical harmonics $Y_\ell^m(\theta, \phi)$ with $\ell = 0, 1, 2, \dots$ and $m = -\ell, \dots, \ell$.
- Radial part $\Rightarrow$ functions $R_{n\ell}(r)$ involving associated Laguerre polynomials.
- Energies quantised as $E_n \propto -1/n^2$, independent of ℓ or m (Coulomb degeneracy).

Solution of the Schrödinger Equation

- Only r and θ appear on the left side and only ϕ appears on the right side of Equation
- The left side of the equation cannot change as ϕ changes.
- The right side cannot change with either r or θ .



- Each side needs to be equal to a constant for the equation to be true.
Set the constant $-m_l^2$ equal to the right side of the equation

- It is convenient to choose a solution to be

Azimuthal equation

$$\frac{\partial^2 g}{\partial \phi^2} = -m_l^2 g e^{im_l \phi}$$

- m_l to be zero or an integer (positive or negative) for this to be true.

Solution of the Schrödinger Equation

- Set the left side of Equation equal to $-m_\ell^2$ and rearrange it.
- Everything depends on r on the left side and ϑ on the right side of the equation.

$$\frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} (E - V) = \frac{m_\ell^2}{\sin^2 \theta} - \frac{1}{f \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right)$$

Solution of the Schrödinger Equation

- Set each side of Equation equal to constant $\ell(\ell + 1)$.

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} \left[E - V - \frac{\hbar^2 \ell(\ell + 1)}{2\mu r^2} \right] R = 0 \quad \text{----Radial equation}$$

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{df}{d\theta} \right) + \left[\ell(\ell + 1) - \frac{m_\ell^2}{\sin^2 \theta} \right] f = 0 \quad \text{----Angular equation}$$

- **Schrödinger equation has been separated into three ordinary second-order differential equations, each containing only one variable.**
- **Radial equation depends on n**
- **Angular equation depends on ℓ**

Complete Solution of the Equations

- The radial wave function R and the spherical harmonics Y determine the probability density for the various quantum states. The total wave function depends on n , ℓ , and m_ℓ . The wave function becomes

$$\psi_{nlm_\ell}(r, \theta, \phi) = R_{nl}(r)Y(\theta, \phi)$$

- Complete solution is the product of the separate solutions
- Total quantum numbers n, ℓ, m_ℓ

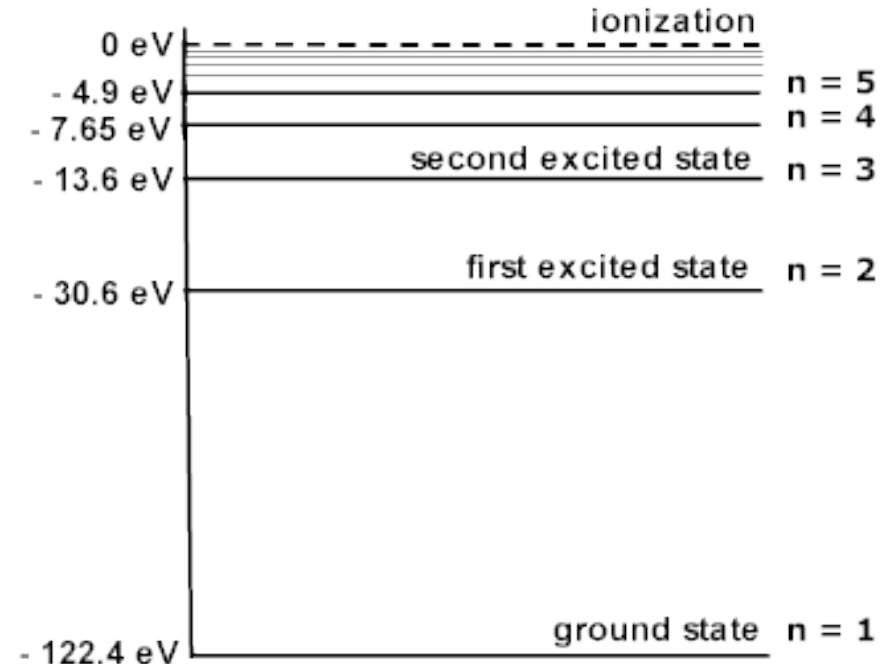
Quantum Numbers

- The appropriate boundary conditions to previous Equation and leads to the following restrictions on the quantum numbers ℓ and m_ℓ :
 - $\ell = 0, 1, 2, 3, \dots$ (degeneracy)
 - $m_\ell = -\ell, -\ell + 1, \dots, -2, -1, 0, 1, 2, \dots, \ell - 1, \ell$ (“orientation”)
 - $|m_\ell| \leq \ell$ and $\ell \geq 0$.
- The predicted energy levels are

$$E_n = -\frac{\mu}{2} \left(\frac{e^2}{4\pi\epsilon_0\hbar} \right)^2 \frac{1}{n^2} = -\frac{E_0}{n^2}$$

$$n = 1, 2, 3, \dots$$

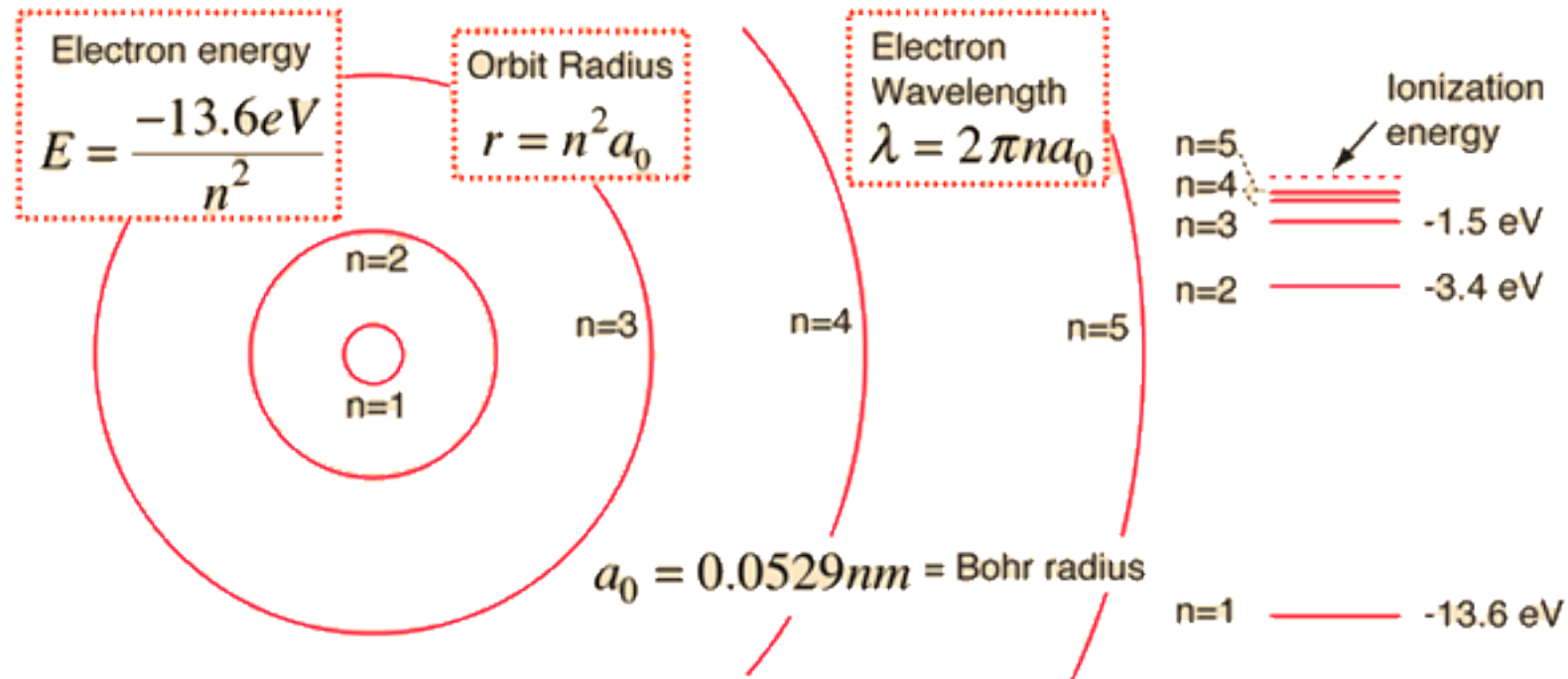
$$n > \ell$$



Solution of the Radial Equation

$$R = Ae^{-r/a_0} \quad a_0 = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2} \quad \left(\frac{1}{a_0^2} + \frac{2\mu}{\hbar^2} E \right) + \left(\frac{2\mu e^2}{4\pi\epsilon_0\hbar^2} - \frac{2}{a_0} \right) \frac{1}{r} = 0$$

$$E = -\frac{\hbar^2}{2\mu a_0^2} = -E_0$$

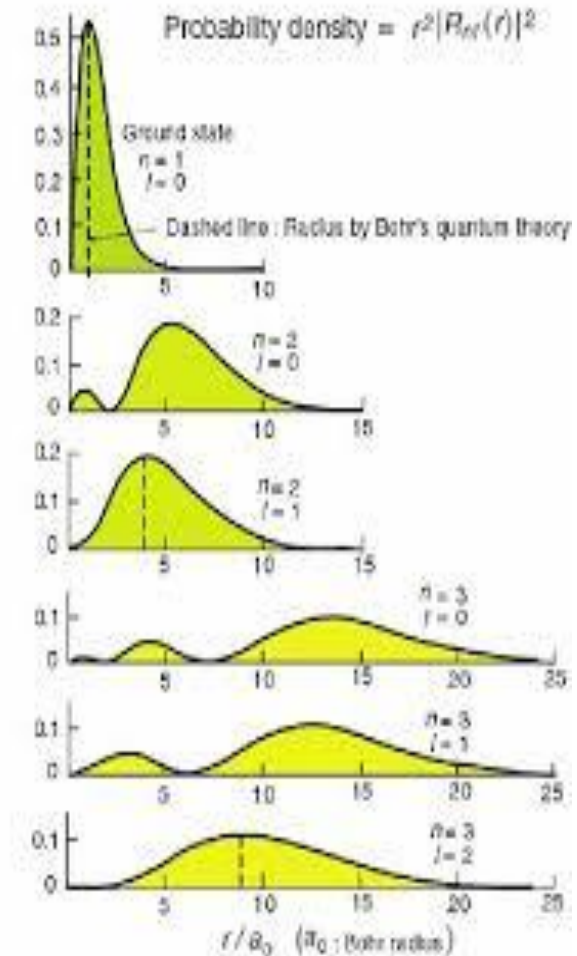


Hydrogen Atom Radial Wave Functions

- First few radial wave functions $R_{n\ell}$

Hydrogen Atom Radial Wave Function

n	ℓ	$R_{n\ell}(r)$
1	0	$\frac{2}{(a_0)^{3/2}} e^{-r/a_0}$
2	0	$\left(2 - \frac{r}{a_0}\right) \frac{e^{-r/2a_0}}{(2a_0)^{3/2}}$
2	1	$\frac{r}{a_0} \frac{e^{-r/2a_0}}{\sqrt{3}(2a_0)^{3/2}}$
3	0	$\frac{1}{(a_0)^{3/2}} \frac{2}{81\sqrt{3}} \left(27 - 18\frac{r}{a_0} + 2\frac{r^2}{a_0^2}\right) e^{-r/3a_0}$
3	1	$\frac{1}{(a_0)^{3/2}} \frac{4}{81\sqrt{6}} \left(6 - \frac{r}{a_0}\right) \frac{r}{a_0} e^{-r/3a_0}$
3	2	$\frac{1}{(a_0)^{3/2}} \frac{4}{81\sqrt{30}} \frac{r^2}{a_0^2} e^{-r/3a_0}$



Solution of the Angular and Azimuthal Equations

- Angular solution of the equation is the product of the two separate solution in the respective variable (angles)
- Group these solutions together into functions.

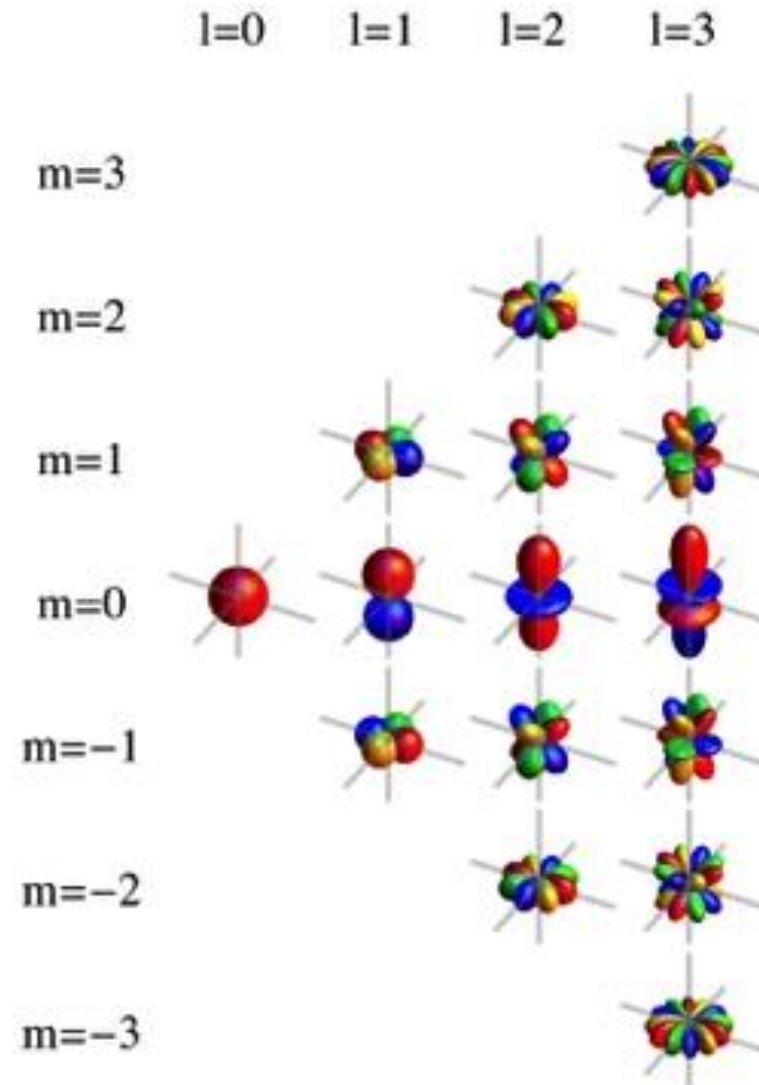
$$Y(\theta, \phi) = f(\theta)g(\phi)$$

---- spherical harmonics

Normalized Spherical Harmonics

Normalized Spherical Harmonics

ℓ	m_ℓ	$Y_{\ell m_\ell}$
0	0	$\frac{1}{2\sqrt{\pi}}$
1	0	$\frac{1}{2}\sqrt{\frac{3}{\pi}} \cos \theta$
1	± 1	$\mp \frac{1}{2}\sqrt{\frac{3}{2\pi}} \sin \theta e^{\pm i\phi}$
2	0	$\frac{1}{4}\sqrt{\frac{5}{\pi}} (3 \cos^2 \theta - 1)$
2	± 1	$\mp \frac{1}{2}\sqrt{\frac{15}{2\pi}} \sin \theta \cos \theta e^{\pm i\phi}$
2	± 2	$\frac{1}{4}\sqrt{\frac{15}{2\pi}} \sin^2 \theta e^{\pm 2i\phi}$
3	0	$\frac{1}{4}\sqrt{\frac{7}{\pi}} (5 \cos^3 \theta - 3 \cos \theta)$
3	± 1	$\mp \frac{1}{8}\sqrt{\frac{21}{\pi}} \sin \theta (5 \cos^2 \theta - 1) e^{\pm i\phi}$
3	± 2	$\frac{1}{4}\sqrt{\frac{105}{2\pi}} \sin^2 \theta \cos \theta e^{\pm 2i\phi}$
3	± 3	$\mp \frac{1}{8}\sqrt{\frac{35}{\pi}} \sin^3 \theta e^{\pm 3i\phi}$



Quantum Numbers

The three quantum numbers:

- n **Principal quantum number (ENERGY LEVEL)**
- ℓ **Orbital angular momentum quantum number (SHAPE)**
- m_ℓ **Magnetic quantum number (ORIENTATION)**

The boundary conditions:

- $n = 1, 2, 3, 4, \dots$ Integer
- $\ell = 0, 1, 2, 3, \dots, n - 1$ Integer
- $m_\ell = -\ell, -\ell + 1, \dots, 0, 1, \dots, \ell - 1, \ell$ Integer

The restrictions for quantum numbers:

- $n > 0$
- $\ell < n$
- $|m_\ell| \leq \ell$

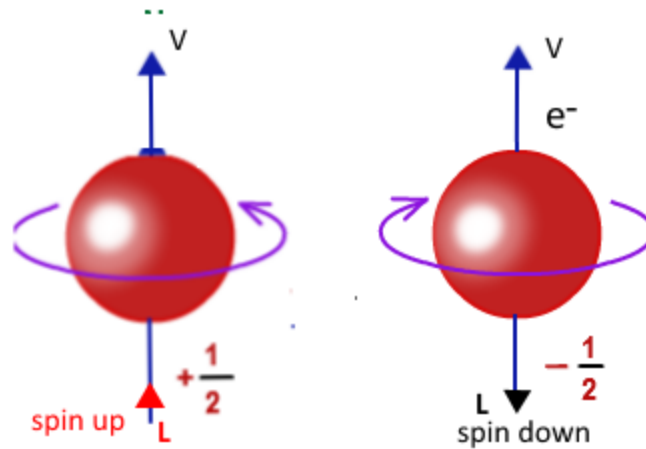
Orbital Angular Momentum Quantum Number ℓ

- A certain energy level is **degenerate** with respect to ℓ when the energy is independent of ℓ .
- Use letter names for the various ℓ values.
 - $\ell = 0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \dots$
 - Letter = $s \quad p \quad d \quad f \quad g \quad h \dots$
- Atomic states are referred to by their n and ℓ .
- A state with $n = 2$ and $\ell = 1$ is called a $2p$ state.
- **The boundary conditions require $n > \ell$.**

Quantum Numbers

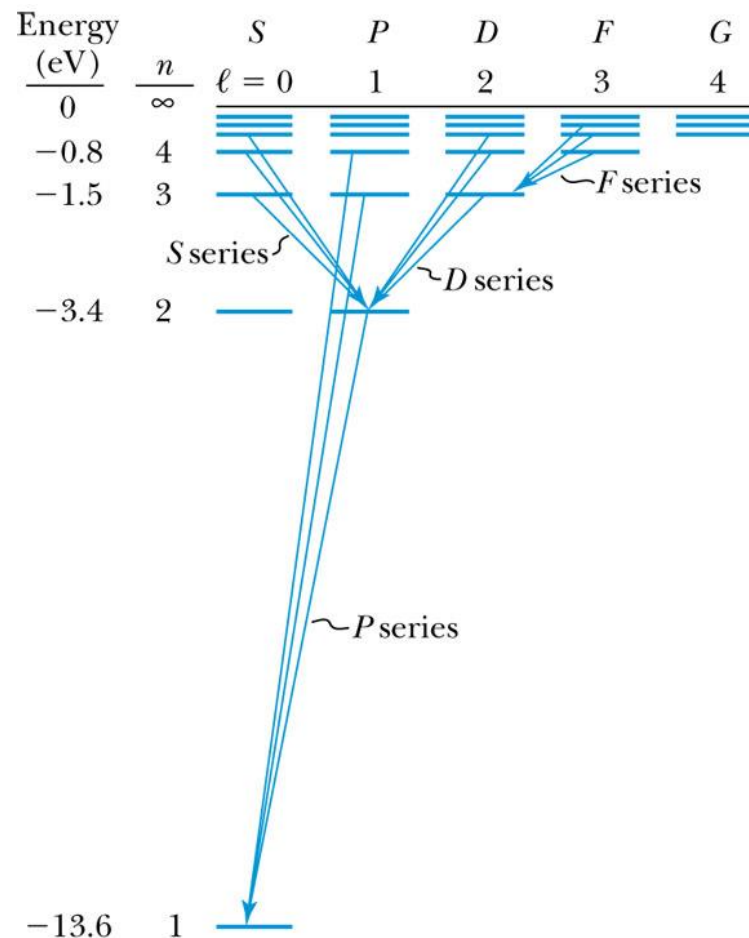
The three quantum numbers:

- n Principal quantum number
- ℓ Orbital angular momentum quantum number
- m_ℓ Magnetic quantum number
- m_s Spin quantum number



Energy Levels and Electron Probabilities

- For hydrogen, the energy level depends on the principle quantum number n .



- In ground state an atom cannot emit radiation. It can absorb electromagnetic radiation, or gain energy through inelastic bombardment by particles.

Selection Rules

- We can use the wave functions to calculate transition probabilities for the electron to change from one state to another.

Allowed transitions:

- Electrons absorbing or emitting photons to change states when $\Delta\ell = \pm 1$.

Forbidden transitions:

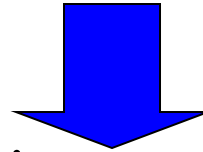
2nd Australian Materials Chemistry Conference
(AMCC25)

- Other transitions possible but occur with much smaller probabilities when $\Delta\ell \neq \pm 1$.

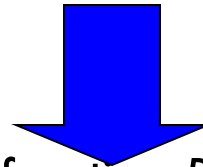
$\Delta n = \text{anything}$ $\Delta\ell = \pm 1$ $\Delta m_\ell = 0, \pm 1$	(Positive integer)
--	--------------------

Probability Distribution Functions

- We must use wave functions to calculate the probability distributions of the electrons.



- The “position” of the electron is spread over space and is not well defined.



- We may use the radial wave function $R(r)$ to calculate radial probability distributions of the electron.
- The probability of finding the electron in a differential volume element $d\tau$ is .

$$dP = \psi^*(r, \theta, \phi) \psi(r, \theta, \phi) d\tau$$

Probability Distribution Functions

- The differential volume element in spherical polar coordinates is

$$d\tau = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

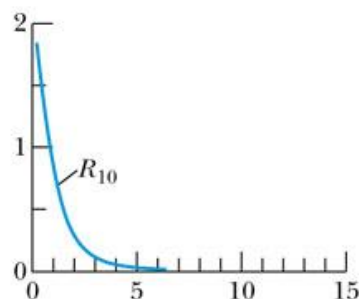
Therefore,

$$P(r) \, dr = r^2 R^* (r) R(r) \, dr \int_0^\pi |f(\theta)|^2 \sin \theta \, d\theta \int_0^{2\pi} |g(\phi)|^2 \, d\phi$$

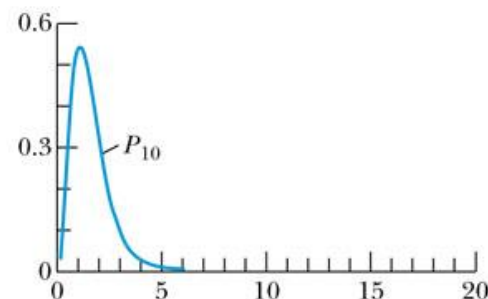
- We are only interested in the radial dependence.
- The radial probability, $P(r) \, dr = r^2 |R(r)|^2 \, dr$, and it depends only on n and l .

Probability Distribution Functions

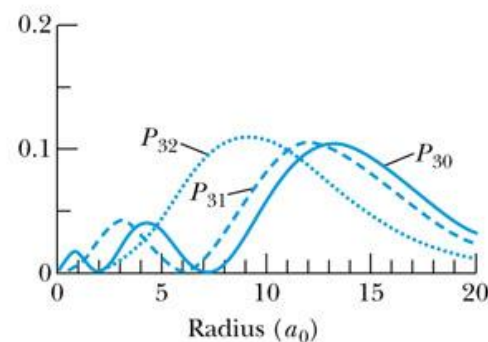
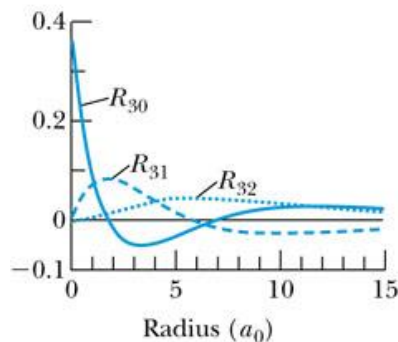
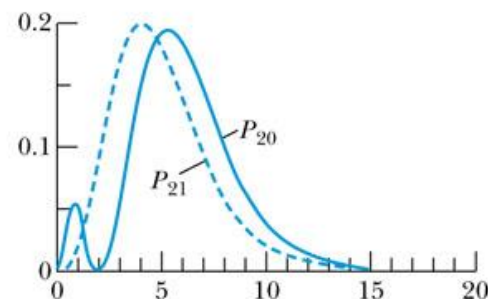
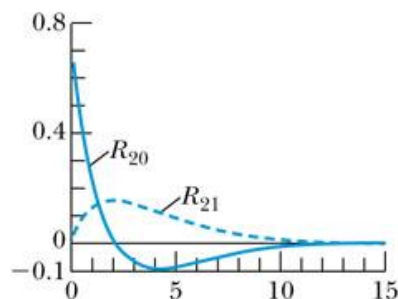
Radial wave functions ($R_{n\ell}$)



Radial probability distribution ($P_{n\ell}$)



- $R(r)$ and $P(r)$ for the lowest-lying states of the hydrogen atom.

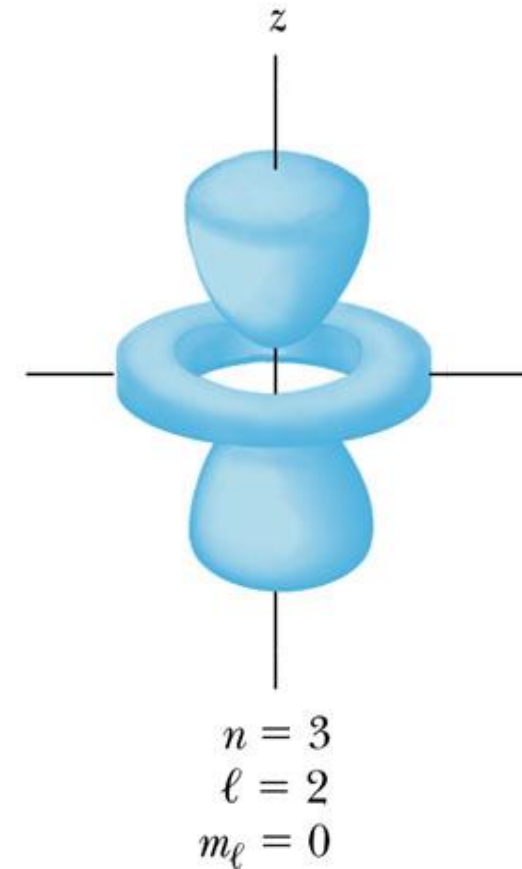
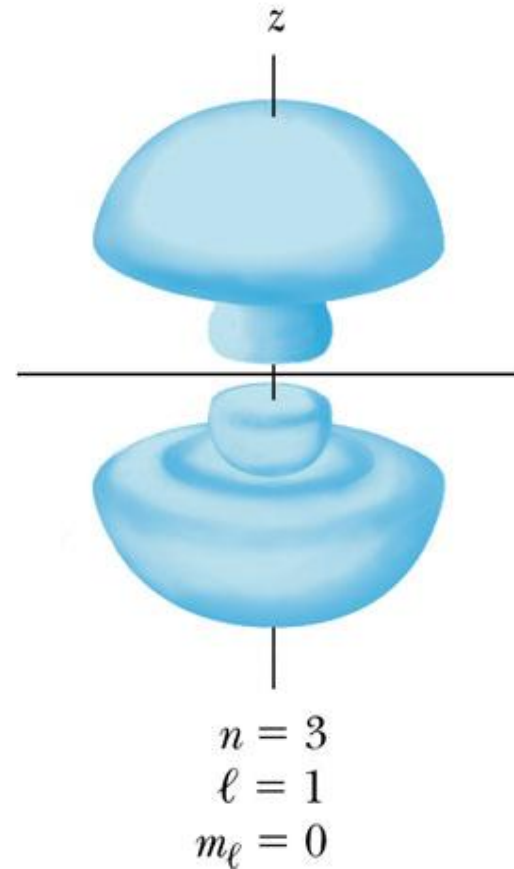
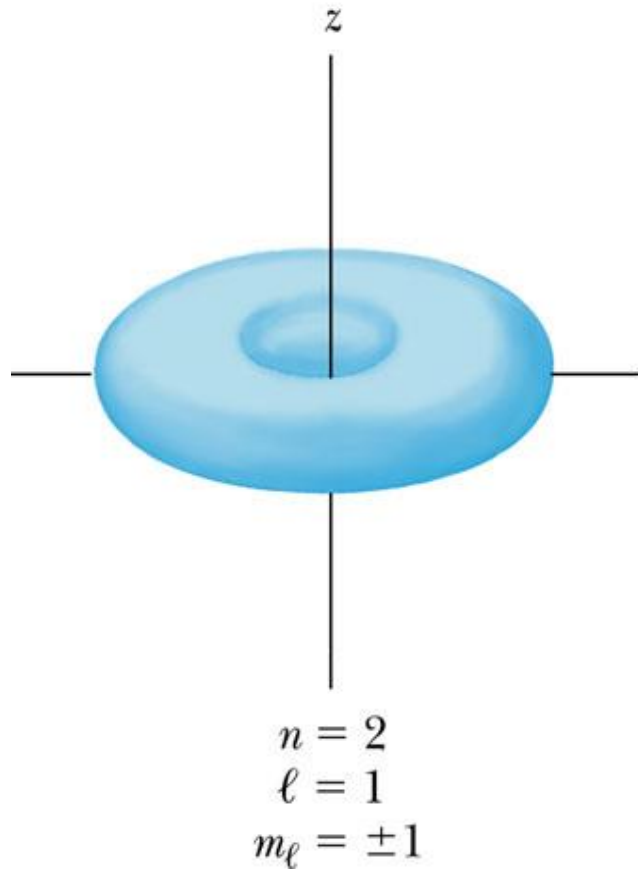


(a)

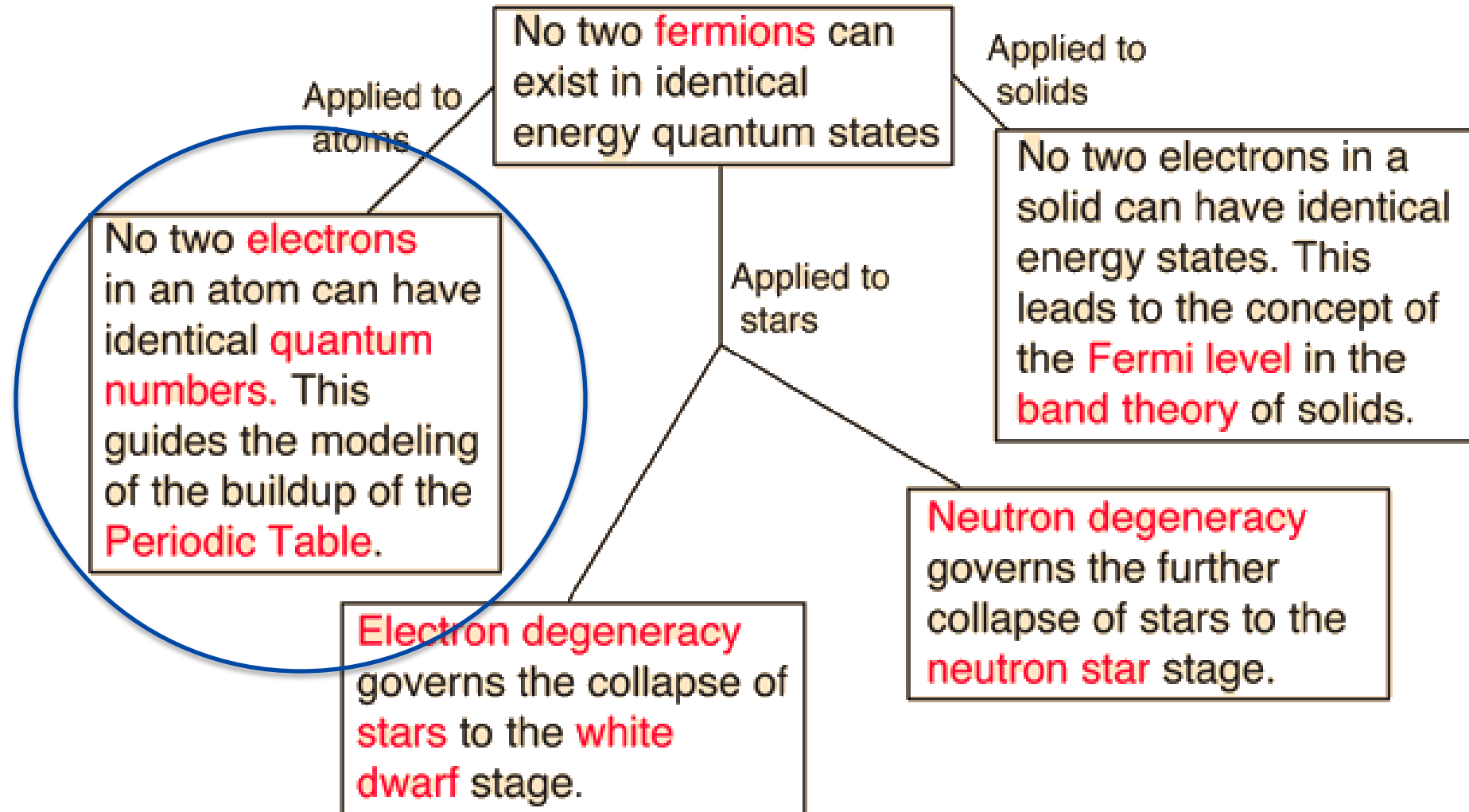
(b)

Probability Distribution Functions

- The probability density for the hydrogen atom for three different electron states.



How orbitals are filled

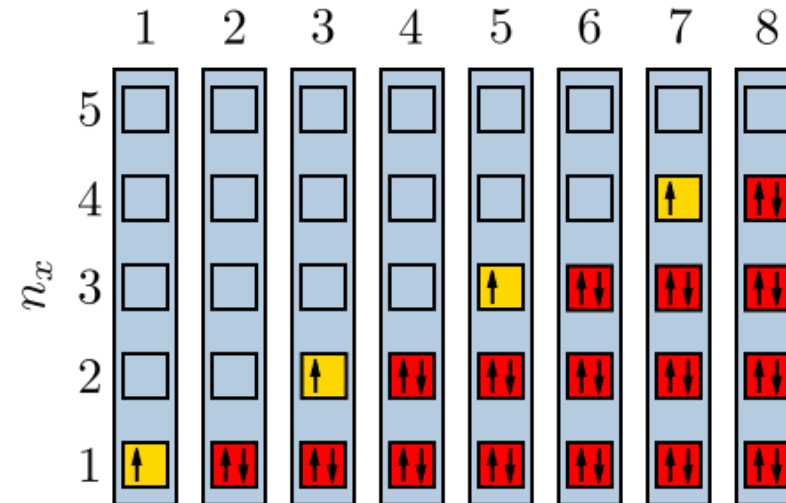


How orbitals are filled

- **Electronic orbitals are filled following some rules:**
- Pauli exclusion principle
- Aufbau rule
- Hund's rule

How orbitals are filled

- Pauli principle implies that only two electron per orbital are allowed
- 2 electrons in $n=1$ because $0 < l < n$, $l < m_l < l$, $m_s = +1/2$ and $-1/2$
- How many electrons in $n=2$?



How orbitals are filled

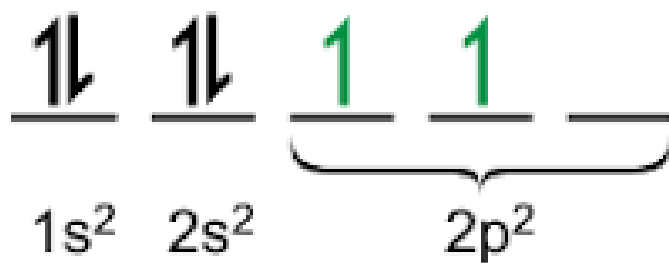
- Aufbau principle
- Electron fill lowest energy level available



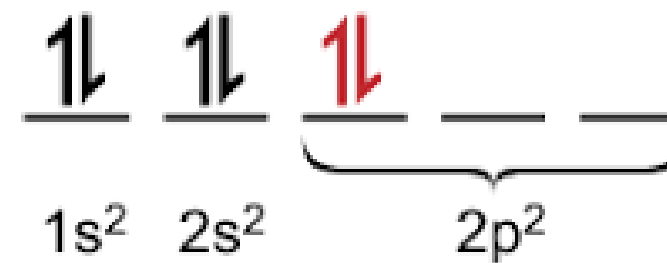
How orbitals are filled

- Hund's rule states that each electrons with same spin must occupy each equal energy orbital before electrons with opposite spin start occupying the same orbital

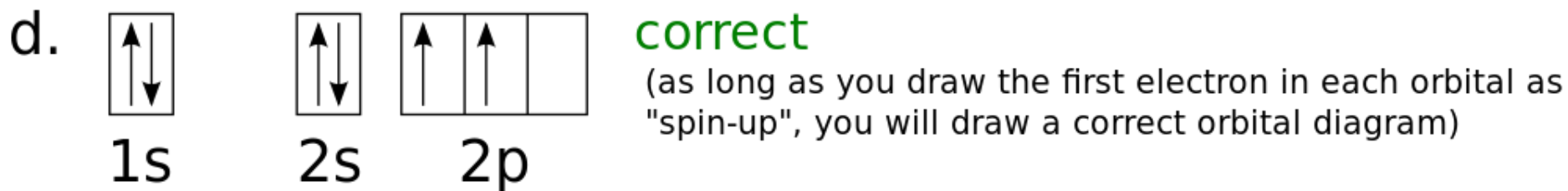
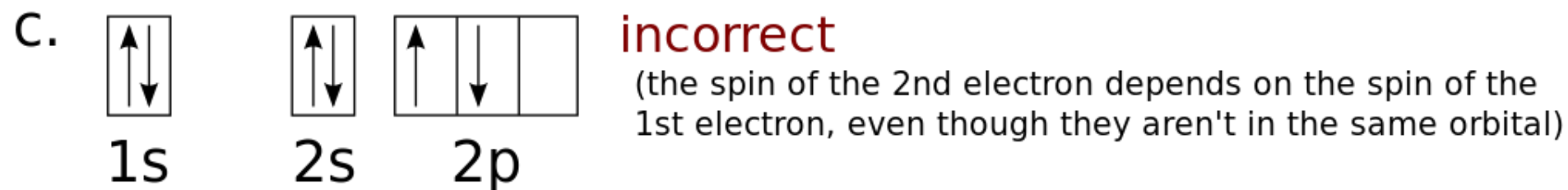
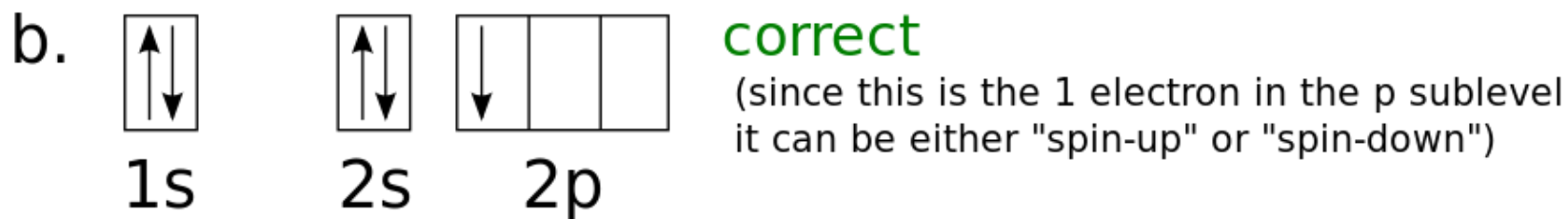
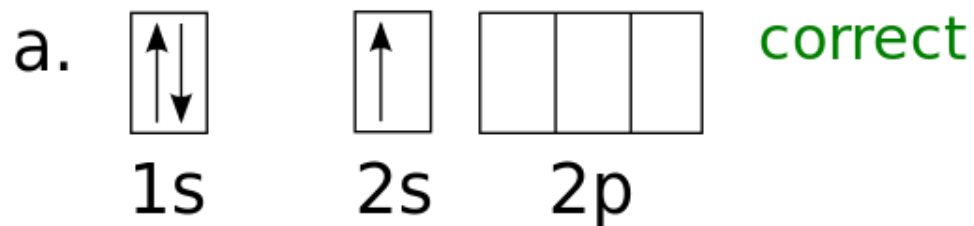
correct



incorrect



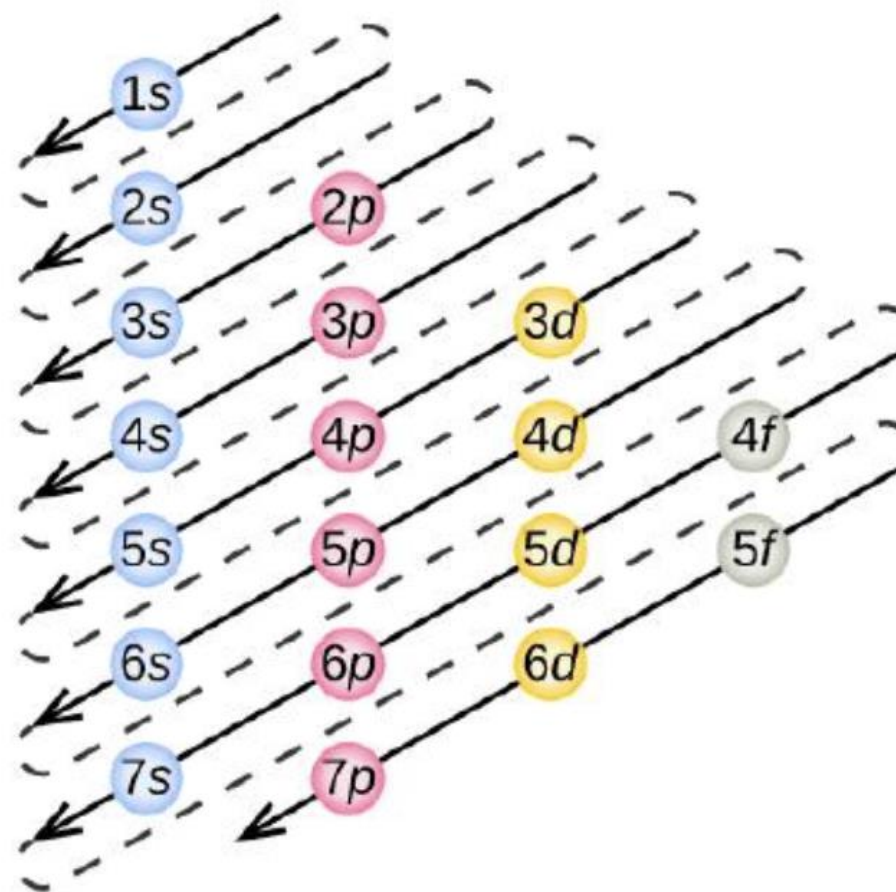
How orbitals are filled



How orbitals are filled

Atom	1s	2s	2p _x	2p _y	2p _z	3s	3p _x	3p _y	3p _z
H	↑								
He	↑↓								
Li	↑↓	↑							
Be	↑↓	↑↓							
B	↑↓	↑↓	↑						
C	↑↓	↑↓	↑	↑					
N	↑↓	↑↓	↑	↑	↑				
O	↑↓	↑↓	↑↓	↑	↑				
F	↑↓	↑↓	↑↓	↑↓	↑				
Ne	↑↓	↑↓	↑↓	↑↓	↑↓				

•Following orbital energy configuration



How orbitals are filled

- **Deviation from fundamental Rules**

- **Pauli**: never violated.

- **Aufbau**: often deviated, especially in transition metals, lanthanides, and heavy atoms.

- Transition metals and heavier elements often have configurations that don't follow the expected order because of **small energy differences** between orbitals.

- **Hund's**: can deviate in cases of strong spin–orbit coupling or crystal-field effects.

- Normally followed, but in certain atoms/molecules, **electron correlation** and **relativistic effects** can cause deviations.

- Example: In some lanthanides and actinides, the balance between exchange energy (favouring Hund's) and spin–orbit coupling can cause unusual arrangements.

Periodic table

IA 1	alkali metals																noble gases										VIIA 18		
H	IIA 2	alkaline earths																post-transition metals										He	
Li	Be	transition metals																III A 13	IV A 14	V A 15	VIA 16	VII A 17	Ne						
Na	Mg	III B 3	IV B 4	V B 5	VI B 6	VII B 7	VIII 8	VIII 9	VIII 10	IB 11	IIB 12	B	C	N	O	F	Ar												
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr												
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe												
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn												
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds																				
																		semimetals (metalloids)										halogens	
Ce Pr Nd Pm Sm Eu Gd Tb Dy Ho Er Tm Yb Lu																		lanthanides											
Th Pa U Np Pu Am Cm Bk Cf Es Fm Md No Lr																		actinides											

S.K. Lower

S.K. Lower

Periodic table

Increased Electronegativity

Increased Ionization Energy (harder to remove an electron)

Decreasing Atomic Radius

Decreased Electronegativity

Decreased Ionization Energy (easier to remove an electron)

Increased Electron Shielding

Increasing Atomic Radius

Increasing number of energy levels

		GROUPS																	
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
PERIODS	1	H He																	
	2	Li Be B C N O F Ne																	
	3	Na Mg Al Si P S Cl Ar																	
	4	K Ca Sc Ti V Cr Mn Fe Co Ni Cu Zn Ga Ge As Se Br Kr																	
	5	Rb Sr Y Zr Nb Mo Tc Ru Rh Pd Ag Cd In Sn Sb Te I Xe																	

More metallic

From *Conceptual Chemistry*, Second Edition by John Suchocki. Copyright © 2004 Benjamin Cummings, a division of Pearson Education.

Alkali Metals

- **All alkali metals have only one electron in their outer shell.**
- **Highly reactive.**
- Low melting points.
- Low densities.
- **Reactive aggressively with halogens to form ionic salts.**
- **React with water to form alkaline hydroxides.**
- Alkali metal (s) + water (aq) → Alkali metal hydroxide (s) + hydrogen (g)
- The vigour of reaction increases down the group to the point that francium has to be stored in mineral oil or kerosene as by itself, it will react with air and literally explode.
- **Have a high electropositivity.**

Alkaline Earths

- shiny.
- silvery-white.
- somewhat reactive metals at standard temperature and pressure.
- **readily lose their two outermost electrons to form cations with a 2+ charge.**
- low densities.
- low melting points.
- low boiling points.

Transition Metals

- have large charge/radius ratio;
- are hard and have high densities;
- have high melting and boiling points;
- form compounds which are often paramagnetic;
- **show variable oxidation states;**
- form coloured ions and compounds;
- **form compounds with profound catalytic activity;**
- **form stable complexes.**

Lanthanides and Actinides

- The lanthanide and actinide series make up the inner transition metals.
- The lanthanide series includes elements 58 to 71, which fill their 4f sublevel progressively.
- The actinides are elements 89 to 103 and fill their 5f sublevel progressively.
- **Lanthanides have different chemistry from transition metals because their 4f orbitals are shielded from the atom's environment.**
- **Actinides are typical metals and have properties of both the d-block and the f-block elements, but they are also radioactive.**

Metalloids

- could be dull or shiny.
- **conducts heat and electricity, but not as well as metals.**
- **good semiconductors.**
- usually malleable.
- usually ductile.
- **can both gain and lose electrons in reactions.**

Halogens

- Halogens are nonmetals in group 17 (or VII) of the periodic table. Down the group, atom size increases. As a diatomic molecule, fluorine has the weakest bond due to repulsion between electrons of the small atoms.
- Due to increased strength of Van der Waals forces down the group, the boiling points of halogens increase. Therefore, the physical state of the elements down the group changes from gaseous fluorine to solid iodine.
- **Due to their high effective nuclear charge, halogens are highly electronegative. Therefore, they are highly reactive and can gain an electron through reaction with other elements. Halogens can be harmful or lethal to biological organisms in sufficient quantities.**

Noble gases

Non chemically
active